

AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY



Department of Computer Science and Engineering

Course No: CSE 2103

Course Title: Data Structures Lab

Assignment No: 01

Date of Submission: 28/06/2026

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Lab Section: A1

Academic Semester: Fall 2025

Program: BSc in CSE

Question : Write a code for implementing Binary Search.

Solution:

Binary Search: Binary Search is an search Algorithm that repeatedly divides the sorted array into halves to find the target element. Works only on sorted data.

Implementation of Binary Search :

```
#include <bits/stdc++.h>
using namespace std;

vector<int> books;
int item; // global variable ok

void binarySearch()
{
    int low = 0;
    int high = books.size() - 1;
    int mid;

    while (low <= high)
    {
        mid = (low + high) / 2;
        if (books[mid] == item)
        {
            cout<<endl<< "Book found";
            return;
        }
        else if (books[mid] < item)
        {
            low = mid + 1;
        }
        else
        {

```

```

        high = mid - 1;
    }
}
cout << "Book not found";
}

void lowerBound()
{
    int low = 0;
    int high = books.size();
    while (low < high)
    {
        int mid = (low + high) / 2;
        if (books[mid] < item)
        {
            low = mid + 1;
        }
        else
        {
            high = mid;
        }
    }
    cout << "The lower bound index of the Book ID is: " <<
low;
}

void upperBound()
{
    int low = 0;
    int high = books.size();
    while (low < high)
    {

```

```

        int mid = (low + high) / 2;
        if (books[mid] <= item)
        {
            low = mid + 1;
        }
        else
        {
            high = mid;
        }
    }
    cout << "The upper bound index of the Book ID is: " <<
low;
}

int main(void)
{
    int size, i, x, temp;
    cout << "Enter the number of Books you want to add: ";
    cin >> size;

    cout << "\nEnter the list of Book ID(s):\n";
    for (i = 0; i < size; i++)
    {
        cin >> x;
        books.push_back(x);
    }

    for (int m = 0; m < size - 1; m++)
    {
        for (int j = 0; j < size - 1 - m; j++)
        {
            if (books[j + 1] < books[j])

```

```
        {
            temp = books[j];
            books[j] = books[j + 1];
            books[j + 1] = temp;
        }
    }
}

cout<<endl<< "Please enter the Book ID you want to
search: ";
cin >> item;

cout << "\nThe following answer is:";
binarySearch();
cout << endl;
lowerBound();
cout << endl;
upperBound();
cout << endl;

return 0;
}
```

Input Samples :

>>Example 1<<

Enter the number of Book you want to add: 5

Enter the list of Book ID(s) 111

222

333

444

555

Please enter the Book ID you want to search: 333

The following answer is:

Book found

The lower bound index of the Book ID is: 2

The upper bound index of the Book ID is: 5

>>Example 2<<

Enter the number of Book you want to add: 3

Enter the list of Book ID(s) 121

123

122

Please enter the Book ID you want to search: 101

The following answer is:

Book not found

The lower bound index of the Book ID is: 0

The upper bound index of the Book ID is: 0